

Wavelet Completeness Is Atomic Regularity

Literature-implied identification for arXiv:2005.04964

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Status: literature-implied answer (exact conceptual characterization). The source's wavelet-complete locally compact groups are exactly the established atomic-regular, or [AR], groups. Equivalently, the Fourier algebra $A(G)$ is a dual Banach space, or $A(G)$ has the Radon–Nikodym property. No new theorem is claimed.

Source question

In Section 7, arXiv PDF page 16, Berge asks to characterize the locally compact groups G satisfying

$$\overline{\bigoplus_{\pi \in \widehat{G}_s} \text{span}_{g \in \mathcal{A}_\pi} \{\mathcal{W}_g f : f \in \mathcal{H}_\pi\}} = L^2(G), \quad (1)$$

where $\widehat{G}_s$ is the set of square-integrable irreducible representations, $\mathcal{A}_\pi$ is the set of admissible vectors modulo phase, and $\mathcal{W}_g f(x) = \langle f, \pi(x)g \rangle$ [1]. The source calls such a group *wavelet complete*.

Identification

Let λ_G be the left regular representation on $L^2(G)$, and let $L^2(G)_{\text{at}}$ denote its atomic part: the closed sum of all irreducible λ_G -invariant subspaces.

Theorem 1. *For a locally compact group G ,*

$$\overline{\bigoplus_{\pi \in \widehat{G}_s} \text{span}_{g \in \mathcal{A}_\pi} \mathcal{W}_g(\mathcal{H}_\pi)} = L^2(G)_{\text{at}}.$$

Consequently the following conditions are equivalent:

1. *G is wavelet complete in the sense of (1);*
2. *λ_G is a Hilbert direct sum of irreducible representations;*
3. *G is an atomic-regular ([AR]) group.*

Under the standard hypotheses of the Fourier-algebra characterization, these are also equivalent to $A(G)$ being a dual Banach space and to $A(G)$ having the Radon–Nikodym property.

Why the theorem is already in the literature. The Duflo–Moore theorem gives, for every square-integrable irreducible π and normalized admissible g , an isometric intertwiner

$$\mathcal{W}_g : \mathcal{H}_\pi \longrightarrow L^2(G), \quad \mathcal{W}_g \pi(y) = \lambda_G(y) \mathcal{W}_g.$$

Its range is therefore an irreducible subrepresentation of λ_G . Conversely, every irreducible subrepresentation of λ_G is square-integrable; the Duflo–Moore discrete-part theorem identifies its intertwining embeddings with coefficient transforms. Varying the admissible vector supplies the multiplicity directions. Thus the closed coefficient sum in (1) is precisely $L^2(G)_{\text{at}}$.

Taylor’s survey states both directions on PDF page 5, then immediately defines an [AR] group to be one whose left regular representation is a direct sum of irreducibles [3]. On PDF page 8 it records the two equivalent Fourier-algebra characterizations in Theorems 6 and 7. Hence Theorem 1 is a direct identification of the 2020 terminology with a class studied since the 1970s, not a new result.

Consequences and scope

The identification sharpens both examples in the source. Compact groups are [AR] by Peter–Weyl, but compactness is not necessary: Taylor gives the real affine group as a noncompact, nonunimodular [AR] example, whose regular representation is a countable direct sum of two square-integrable irreducibles. The reduced Heisenberg groups have a nonzero continuous regular part, agreeing with the source’s negative result.

For second-countable unimodular type-I groups, the same statement reads particularly concretely:

$$G \text{ is wavelet complete} \iff \text{Plancherel measure is purely atomic.}$$

Each Plancherel atom is a discrete-series representation, and its matrix coefficients span the corresponding Hilbert–Schmidt fiber.

This is an exact representation-theoretic and Banach-space characterization. It is not a simple structural enumeration of every locally compact [AR] group; that older classification program remains nontrivial. The separate global HRT conjecture in the source is unaffected and remains open.

Search and provenance

The run indexes were searched for arXiv:2005.04964, HRT, wavelet completeness, Plancherel atoms, and atomic regularity. Targeted searches for “wavelet complete”, “completely reducible regular representation”, “AR-group”, and “purely atomic Plancherel measure” located the older [AR] literature. Taylor’s 2008 chapter predates arXiv:2005.04964 and therefore does not explicitly claim to answer Berge’s question. The implication is agent-identified, so the correct provenance is *literature implied*, not *literature already answered*.

References

- [1] E. Berge, *Interpolation in Wavelet Spaces and the HRT-Conjecture*, J. Fourier Anal. Appl. **26** (2020), article 72; arXiv:2005.04964.
- [2] M. Duflo and C. C. Moore, *On the regular representation of a nonunimodular locally compact group*, J. Funct. Anal. **21** (1976), 209–243.
- [3] K. F. Taylor, *Groups with Atomic Regular Representation*, in *Representations, Wavelets, and Frames*, Applied and Numerical Harmonic Analysis, Birkhäuser Boston (2008), 33–45; doi:10.1007/978-0-8176-4683-7_3.